WinNovation

NEP 2020 & TECHNICAL EDUCATION SKILL ELECTIVE

NATIONAL EDUCATIONAL BLUEPRINT • 44-DAY EXPERIENTIAL TRACK

APPLIED DATA ANALYTICS & BUSINESS INTELLIGENCE

A comprehensive, industry-aligned STEM & vocational curriculum engineered for Colleges, Polytechnics, Skill Hubs & Advanced Technology Institutes.



DURATION
44 Days
88 Instructional Periods (2 hrs/day)

PEDAGOGICAL SPLIT
70 : 30
70% Hands-on Lab : 30% Theory

TARGET STANDARD
Undergraduate & Diploma
NSQF Level 5/6 Aligned

INDUSTRY DELIVERABLE
20+ Projects
5 Capstones + 15 Mini-Projects

Winnovation Education Systems • Empowering Tomorrow's Analytical & Engineering Leaders

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Academic Year 2026–27 | Reference Document Specification

Campus: 1157, i-Thum Galleria, Alpha 2, Greater Noida, UP 201308

1. Pedagogical Vision & NEP 2020 Strategic Framework

The **Winnovation Applied Data Analytics Curriculum** transitions undergraduate learners and technology students from passive consumers of technology to proactive data storytellers and diagnostic problem solvers. Built strictly on **Competency-Based Education (CBE)** principles, the curriculum integrates commercial mathematics, statistics, computer science, and business decision-making through experiential case studies.

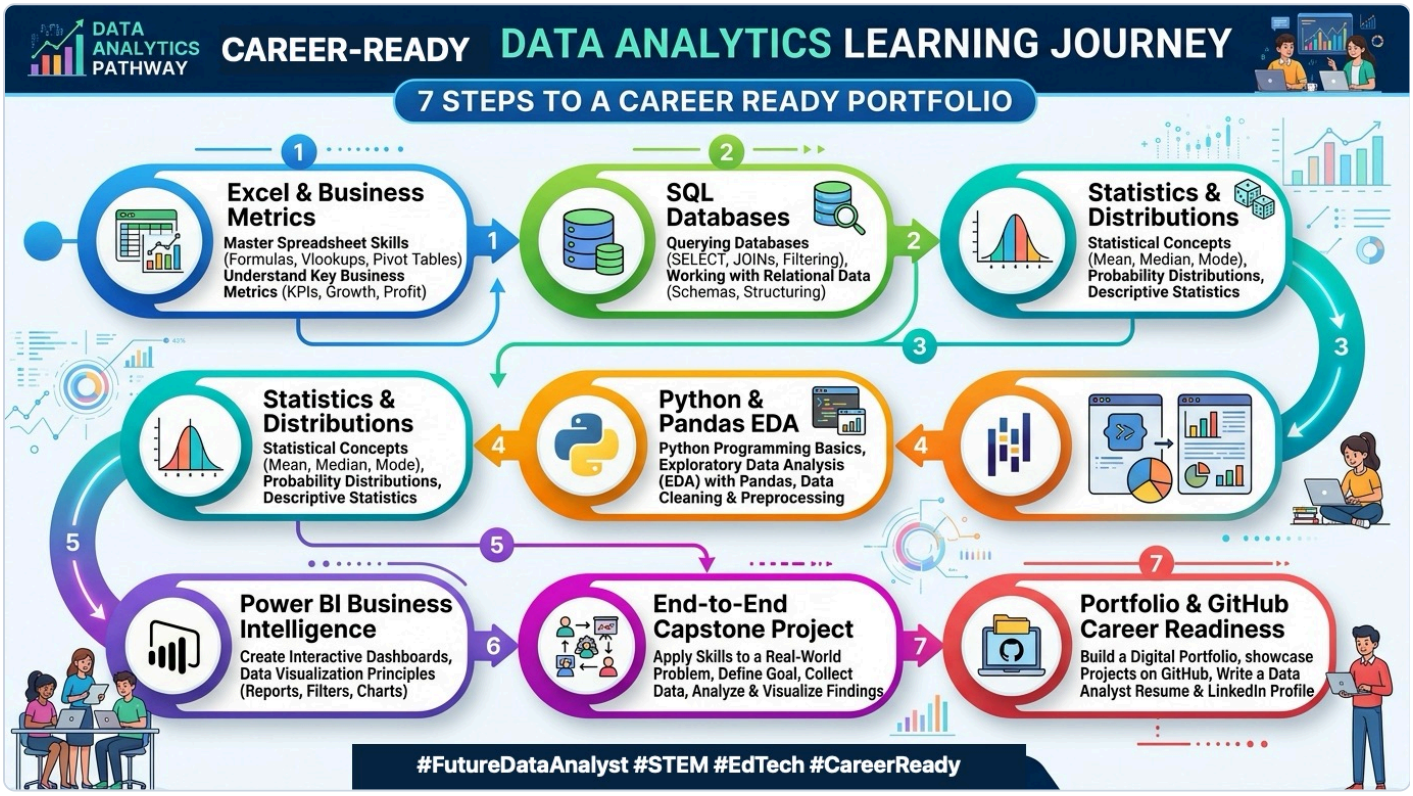
Core Professional Competencies

- **Diagnostic Reasoning:** Auditing anomalies and finding operational root-causes.
- **Quantitative Fluency:** KPI modeling, probability, and hypothesis testing.
- **Modern Tool Proficiency:** Excel, SQL, Python, Pandas, Power BI, GitHub.

Indian Industry & Socio-Economic Context

- Real datasets from UPI digital transactions, ONDC e-commerce, and logistics.
- Public health insights modeled on Ayushman Bharat operational data.
- Strict adherence to the **Digital Personal Data Protection (DPDP) Act, 2023**.

2. 7-Stage Experiential Learning Pathway



3. Examination & Assessment Rubric (Institutional Vocational Pattern)

EVALUATION PILLAR	ASSESSMENT INSTRUMENT	WEIGHTAGE	EVALUATION BENCHMARKS & STANDARDS
Continuous Formative Lab	15+ Daily Mini-Project Deliverables	**30 Marks**	Data cleaning audits, formula precision, SQL query hygiene, error handling
Mid-Term Milestone Capstones	Excel Dashboard (Day 9) + SQL Capstone (Day 18)	**20 Marks**	Multi-dimensional data models, query optimization, executive UI/UX
Comprehensive End-Term Capstone	Healthcare Analytics Ecosystem (Days 39-43)	**20 Marks**	Charter, SQL pipeline, Python EDA notebook, Power BI report & executive defense
Comprehensive Practical Lab Examination	Certified Lab Journal & Live Lab Problem	**15 Marks**	Invigilated hands-on query writing, dashboard generation, verified lab file
Viva Voce & Technical Defense	Oral Examination & Business Communication	**15 Marks**	Interpretation of business KPIs, root-cause storytelling, analytical reasoning
TOTAL QUALIFYING AGGREGATE		**100 Marks**	**Minimum Qualifying Benchmark: 40% in Theory & 40% in Practical**

PHASE 1: SPREADSHEET MODELLING & BUSINESS METRICS (DAYS 1 – 9)

Tool: Microsoft Excel / LibreOffice Calc

DAY	CORE THEORETICAL TOPICS	GUIDED LABORATORY PRACTICAL	MINI PROJECT & DELIVERABLE
Day 01	**Introduction to Data Analytics** • Role: Data Analyst vs. Data Scientist vs. BI • Descriptive, Diagnostic, Predictive, Prescriptive • Analytics Lifecycle; Business vs. Tech problems	Load Indian retail sales records. Isolate categorical dimensions from numeric measures; formulate 5 business inquiries.	Mini Project **Business Problem Identification** Problem Statement + 5 KPIs + 5 Questions.
Day 02	**KPIs & Business Metrics** • Revenue, COGS, Gross Profit, Margin % • Orders, Customers, Average Order Value (AOV) • Conversion Rate, Retention, Churn %, Growth %	Calculate Margin % and Churn % on 1,000-order retail transactions sheet using arithmetic formulas.	Mini Project **Retail KPI Analysis** Tabular KPI calculation sheet with business interpretation note.
Day 03	**Data Quality & Cleaning Concepts** • 6 Quality Dimensions; Missing values • Duplicate records, incorrect types, outliers • Inconsistent categories, validation, data dictionary	Audit an unformatted dataset: detect null IDs, corrupt dates, and regional spelling variants ("BLR", "Bangalore").	Mini Project **Data Quality Audit** Institutional audit log listing error types, counts, and risks.
Day 04	**Excel Fundamentals & Structure** • Interface, rows/columns/cells, data types • Structured Tables (`Ctrl+T`), Sorting, Filters • Freeze Panes, Data Validation, References (`$A$1`)	Construct a normalized Sales Master sheet. Apply Data Validation dropdowns for Indian States & Payment Modes (UPI, Cards).	Mini Project **Sales Data Preparation** Clean, error-resistant structured Excel table with active validation controls.
Day 05	**Mathematical & Logical Formulas** • `SUM` , `AVERAGE` , `COUNT` , `COUNTA` , `MIN` , `MAX` , `MEDIAN` • Logical: `IF` , `AND` , `OR` , `IFERROR` • Defensive spreadsheet design	Calculate Revenue, Margin, Commission Tiers, and Sales Quota statuses based on compound business logic.	Mini Project **Sales Performance Calculator** Automated Excel analysis sheet computing rep-wise performance.
Day 06	**Relational Lookup Functions** • Modern `XLOOKUP` vs. legacy `VLOOKUP` • `INDEX` + `MATCH` two-way dynamic search • Exact vs. Approx match; Error handling; Merging	Merge three separate organizational sheets: Employee Data, Department Roster, and Compensation Structure.	Mini Project **Employee Information Analyzer** Consolidated employee dataset with dynamic compensation lookup logic.
Day 07	**Systematic Data Cleaning in Excel** • Duplicates, Find & Replace • Text: `TRIM` , `CLEAN` , `LEFT` , `RIGHT` , `MID` , `LEN` , `PROPER` • Text to Columns, date format normalization	Clean an unformatted CRM file: sanitize phone numbers, split addresses, standardize names into Proper Case.	Mini Project **Customer Data Cleaning Project** Side-by-side Before/After worksheets + reproducible cleaning audit log.
Day 08	**Pivot Tables & Interactive Charts** • Pivot Tables: Rows, Columns, Values, Filters • Summarization: Running totals, % of Total • Pivot Charts, Slicers, Timelines, visual rules	Analyze sales by Indian region, category, and month. Configure synchronized Slicers across multiple Pivot tables.	Mini Project **Retail Performance Report** Multi-pivot analytical sheet featuring interactive slicers and dynamic charts.

PHASE 1 MILESTONE CAPSTONE • DAY 09

Executive Sales Performance Dashboard Project

Core Inquiries: (1) Which zone generates maximum revenue? (2) What is the monthly trajectory? (3) Top 10% reps? (4) Margin erosion analysis.
Architecture: Strategic KPI cards (Revenue ₹, Profit ₹, Orders, AOV) + Dual-axis Monthly Trend Line + Zonal Heatmap + Region/Category Slicers.
Deliverable: Professional Excel dashboard workbook (`.xlsx`) + 2-page Executive Insights & Action Memo.

PHASE 2: RELATIONAL DATABASES & SQL (DAYS 10 – 18)

Environment: MySQL 8.0 / PostgreSQL

DAY	CORE THEORETICAL TOPICS	GUIDED LABORATORY PRACTICAL	MINI PROJECT & DELIVERABLE
Day 10	**Database & SQL Fundamentals** • RDBMS: Tables, Records, Fields, PK, FK • Entity Relationships: 1:1, 1:Many, Many:Many • DDL: `CREATE`, `ALTER`, `DROP TABLE`; Constraints	Build an E-Commerce database: `Customers`, `Products`, `Orders` tables with active foreign key constraints.	Mini Project **Build an E-Commerce Database** Production SQL DDL script creating tables and relational constraints.
Day 11	**SELECT Queries & Filtering** • `SELECT`, `DISTINCT`, Aliases (`AS`) • `WHERE`, `AND`, `OR`, `NOT`, `IN`, `BETWEEN` • Pattern matching: `LIKE`; handling `NULL`	Query orders shipped to Delhi or Mumbai with order totals > ₹5,000. Identify profiles missing emails.	Mini Project **Customer Search & Analysis** Documented SQL script answering 20 granular operational inquiries.
Day 12	**Sorting & Aggregations** • `ORDER BY` (ASC/DESC), `LIMIT` / `TOP` • Aggregates: `COUNT(*)`, `SUM`, `AVG`, `MIN`, `MAX`	Calculate enterprise gross revenue, average transaction value, highest individual order, and lowest ticket size.	Mini Project **Sales KPI Analysis using SQL** Verified SQL script computing foundational enterprise metrics.
Day 13	**Grouped Aggregations & HAVING** • `GROUP BY` single & multiple columns • `HAVING` mechanics vs. `WHERE` • Conditional aggregations & cohort analysis	Compute revenue and order volume by Category and Region. Filter groups having total revenue > ₹5,00,000.	Mini Project **Regional Sales Analysis** Grouped SQL script producing performance matrices across fulfillment zones.
Day 14	**Relational SQL JOINs** • `INNER`, `LEFT`, `RIGHT`, `FULL JOIN` • Primary & foreign key linkage; Multi-joins • Detecting orphaned records & duplicate rows	Execute a 4-table join: `Customers` → `Orders` → `Order_Items` → `Products` to build unified transaction summaries.	Mini Project **Customer & Product Analysis** Production SQL join queries extracting consolidated multi-table records.
Day 15	**CASE WHEN & NULL Handling** • `CASE WHEN ... THEN ... ELSE ... END` • Null imputation: `COALESCE()`, `IFNULL()` • Discretization & customer tier segmentation	Segment customers into VIP (> ₹50k), Gold (₹15k–₹50k), and Standard (< ₹15k) spending tiers. Replace null phone numbers.	Mini Project **Customer Segmentation using SQL** Dynamic segmentation script classifying customers and product margins.
Day 16	**Subqueries & Common Table Expressions** • Scalar, nested, and correlated subqueries • Common Table Expressions (CTEs) using `WITH` • Query readability, modularity & performance	Find customers spending above the national average. Build a multi-CTE pipeline benchmarking SKUs against category averages.	Mini Project **Business Performance Analysis** Modular CTE-structured SQL script resolving multi-level analytical questions.
Day 17	**Enterprise Window Functions** • `OVER()`, `PARTITION BY`, `ORDER BY` • Ranking: `ROW_NUMBER`, `RANK`, `DENSE_RANK` • Offsets: `LAG`, `LEAD`; Running totals; % contribution	Rank top 3 products in each category using `DENSE_RANK()`. Compute Month-over-Month (MoM) revenue growth using `LAG()`.	Mini Project **Customer Ranking & Growth Analysis** Window function script computing cumulative sales and purchase sequences.

PHASE 2 MILESTONE CAPSTONE • DAY 18

SQL Capstone: E-Commerce Analytics Project

Core Inquiries: (1) Top 1% power customers? (2) Best-selling products? (3) Top category? (4) Monthly growth? (5) Repeat rate? (6) Declining SKUs? (7) Revenue contribution %.
Deliverables: Production SQL script (`.sql`) + Comprehensive KPI Report + Business Insights Memo + GitHub Markdown Documentation.

PHASE 3: BUSINESS STATISTICS & QUANTITATIVE METHODS (DAYS 19 – 22)

Mathematical Foundations

DAY	CORE THEORETICAL TOPICS	GUIDED LABORATORY PRACTICAL	MINI PROJECT & DELIVERABLE
Day 19	**Descriptive Statistics**: Mean, Median, Mode, Range, Quartiles, Percentiles, Interquartile Range (IQR), Tukey 1.5 x IQR Outlier Rule.	Compute five-number summaries for retail customer ticket sizes. Flag orders beyond upper fence as anomalies.	Mini Project **Sales Distribution Analysis**: Statistical dispersion report with outlier thresholds.
Day 20	**Variance & Distribution**: Variance (σ^2), Standard Deviation (σ), Normal Gaussian Curve, Empirical 68–95–99.7% Rule, Skewness, Z-Scores.	Standardize daily store revenues into Z-scores: $Z = (X - \mu) / \sigma$. Identify days with $\lvert Z \rvert > 3$ as volatility spikes.	Mini Project **Sales Risk & Outlier Analysis**: Statistical audit modeling revenue stability.
Day 21	**Correlation & Regression**: Pearson's r, Positive/Negative, Correlation vs. Causation, Simple Linear Regression ($Y = \beta_0 + \beta_1 X$), R^2 fit.	Analyze digital ad spend against sales for a D2C Indian brand. Compute r and forecast sales for a ₹50,000 budget increase.	Mini Project **Marketing Spend vs Sales Analysis**: Regression workbook with scatter plots and forecasts.
Day 22	**Hypothesis Testing & A/B Testing**: H_0 vs. H_1, p-values, 95% Confidence Intervals, Type I/II errors, Two-sample t-tests, A/B methodology.	Test if redesigning a UPI payment button ("Pay Now" vs. "Complete Payment") caused a statistically significant lift in conversion.	Mini Project **Marketing Campaign A/B Test**: Complete experimentation report with p-value analysis.

PHASE 4: COMPUTATIONAL ANALYTICS WITH PYTHON (DAYS 23 – 29)

Python 3.11 / Pandas / NumPy / Matplotlib

DAY	CORE THEORETICAL TOPICS	GUIDED LABORATORY PRACTICAL	MINI PROJECT & DELIVERABLE
Day 23	**Python Fundamentals**: Variables, Data types, Operators, Strings, Lists, Tuples, Dictionaries, Conditions (`if-elif-else`), Loops.	Build a commercial business calculator: compute GST (18%), tiered discounts, and invoice billings.	Mini Project **Business Calculator**: Documented Python script (`.py` / `.ipynb`) processing retail rules.
Day 24	**Functions & Data Structures**: Parameters, return values, List methods, Dictionary operations, List comprehensions, Modules (`math`, `datetime`).	Write modular functions for financial metrics (`calculate_margin`). Process orders using list comprehensions.	Mini Project **Revenue & Profit Automation**: Modular Python script with automated business functions.
Day 25	**NumPy Array Computing**: Arrays, Shape, Indexing, Slicing, Vectorization, Mathematical operations, Statistical functions on matrices.	Simulate a 365-day array of daily sales across 5 branches. Compute branch-wise mean, std, and 90th percentile without loops.	Mini Project **Financial Data Analysis**: Vectorized array simulation modeling multi-store sales.
Day 26	**Pandas Tabular Engineering**: Series, DataFrames, `read_csv`, `head`, `info`, `describe`, Filtering, Sorting, `groupby`, Aggregation, Merge.	Ingest retail transactions. Group by customer city, compute ticket sizes, and merge customer profile records with orders.	Mini Project **Retail Data Analysis**: Pandas data manipulation script extracting business summaries.
Day 27	**Python Data Cleaning**: Missing values (`isna`, `fillna`, `dropna`), duplicates, datatype casting, string cleaning (`.str`), date conversion, feature engineering.	Clean an Indian customer dataset: parse dates, impute income using zonal medians, derive `tenure_days`.	Mini Project **Customer Dataset Cleaning**: Automated preprocessing pipeline outputting clean CSV.
Day 28	**Exploratory Data Analysis (EDA)**: Univariate, bivariate, multivariate analysis; Distribution, Outliers, Trends, Segmentation, Correlation matrices (`.corr`).	Perform EDA on a 5,000-row retail loan dataset. Detect correlations between applicant income, credit score, and default rates.	Mini Project **Exploratory Sales Analysis**: Deep-dive EDA notebook with correlation heatmaps and profiles.

PHASE 4 MILESTONE CAPSTONE • DAY 29

Python Visualization & Customer Behavior Analytics Project

Scope: Matplotlib charts (Bar, Line, Histogram, Scatter), Demographics, Purchase frequency, Revenue contribution, Repeat buyers, RFM customer tiers.
Deliverable: Production Jupyter Notebook (`.ipynb`) + Publication-grade visualizations + Strategic retention recommendations.

PHASE 5: POWER BI & DATA MODELING (DAYS 30 – 38)

Tool: Microsoft Power BI Desktop

DAY	CORE THEORETICAL TOPICS	GUIDED LABORATORY PRACTICAL	MINI PROJECT & DELIVERABLE
Day 30	**Power BI Fundamentals**: Desktop architecture (Report, Table, Model views), Ingestion of Excel/CSV, Datatypes, Visuals palette, Filters, Canvas themes.	Import annual sales records. Configure page dimensions, theme palette, and build a single-page overview canvas with KPI cards.	Mini Project **Business Performance Report**: Clean single-page visual overview (`.pbix`).
Day 31	**Power Query ETL**: Power Query Editor, M language steps, Remove columns/rows, Replace values, Change data types, Split columns, Merge columns.	Clean an unformatted sales export: split compound invoice codes, trim whitespace, replace null discounts with `0` .	Mini Project **ETL Data Cleaning Project**: Repeatable Power Query transformation recipe.
Day 32	**Advanced Power Query**: Merge queries (Joins), Append queries (Unions), Unpivot columns (matrix to tabular), Conditional columns, Custom columns, Multi-file combining.	Consolidate 12 monthly branch sales files into one table. Unpivot quarterly forecast columns into a normalized schema.	Mini Project **Multi-File Sales ETL Project**: Automated multi-source ingestion pipeline.
Day 33	**Relational Data Modeling**: Fact tables vs. Dimension tables, Star schema vs. Snowflake, Relationships, Cardinality (1 : Many, 1 : 1, Many : Many), Cross-filter direction, Date Dimension.	Deconstruct flat sheet into `Fact_Sales` , `Dim_Customer` , `Dim_Product` , `Dim_Store` , `Dim_Date` . Set 1 : Many links.	Mini Project **Retail Star Schema Project**: Fully normalized relational model verified in Model View.
Day 34	**DAX Fundamentals**: Calculated columns (Row Context) vs. Explicit Measures (Filter Context), `SUM` , `COUNT` , `DISTINCTCOUNT` , `DIVIDE` , `CALCULATE` intro.	Author explicit measures: `[Total Revenue]` , `[Total Cost]` , `[Gross Profit]` , `[Margin %]` . Build a dedicated measures table.	Mini Project **Business KPI Measure Project**: Certified DAX metrics cataloged in a structured model.
Day 35	**Advanced DAX & Filter Context**: Deep dive into `CALCULATE` , Modifier functions: `FILTER` , `ALL` , `ALLEXCEPT` , `REMOVEFILTERS` , Dynamic KPIs, % of Total.	Calculate National Benchmark Revenue using `ALL(Dim_Store)` and derive `[% Regional Contribution]` .	Mini Project **Profitability Analysis using DAX**: Context-aware financial measures suite.
Day 36	**Time Intelligence in DAX**: Continuous Date table requirements, `TOTALYTD` , `TOTALMTD` , `TOTALQTD` , `SAMEPERIODLASTYEAR` , YoY growth, MoM growth, Rolling averages.	Build time measures: `[Revenue YTD]` , `[Revenue Prior Year]` , and calculate dynamic Year-over-Year (YoY) Growth %.	Mini Project **Sales Growth Analysis**: Interactive multi-period time intelligence reporting page.
Day 37	**Dashboard UX & Visual Storytelling**: Layout hierarchy (F-pattern), Visual contrast, KPI cards, Slicers, Drill-through pages, Visual Tooltips, Bookmarks, Conditional formatting.	Configure Bookmarks to toggle between "Sales View" and "Profit View". Design custom hover tooltip cards for granular customer details.	Mini Project **Executive Dashboard Challenge**: UX-audited reporting canvas meeting corporate standards.

PHASE 5 MILESTONE CAPSTONE • DAY 38

Power BI Milestone: Executive Sales Intelligence Dashboard

3-Page Architecture: *Page 1:* Executive Overview (Revenue, Profit, Orders, AOV, Trajectory, Bullet charts) • *Page 2:* Product & Customer (Top/Bottom SKUs, 80/20 Pareto, Profitability matrix) • *Page 3:* Regional Performance (Zonal maps, Salesperson quota rankings, Monthly trends).
Deliverable: Fully interactive Power BI Dashboard (`.pbix`) + Executive PDF report + 5-minute live demonstration.

PHASE 6: END-TO-END MULTI-DISCIPLINARY CAPSTONE (DAYS 39 – 43) Scenario: Healthcare Systems & Patient Analytics

Students act as Lead Healthcare Analysts for an Indian multi-specialty hospital network managing 5 relational tables: `Patients`, `Doctors`, `Appointments`, `Billing`, and `Departments`.

STAGE / DAY	INSTRUCTIONAL SCOPE & CORE ACTIVITY	GUIDED EXECUTION TASKS	CAPSTONE DELIVERABLE
Day 39 Inception	Stakeholder requirements, Business objectives, Data dictionary, KPI identification, Data sources, Data-quality audit.	Translate hospital board goals into technical metrics: OPD wait times, bed occupancy, doctor utilization, no-show rates.	**Healthcare Project Charter** Formal scope document + Data Dictionary + DPDP Act governance protocol.
Day 40 SQL Pipeline	SQL extraction, Joins, Aggregations, CTEs, Window functions.	Extract multi-table records: Analyze Patient volume, Department performance, Appointment trends, Revenue, Cancellation rate.	**Healthcare SQL Pipeline** Production SQL script extracting validated analytical view tables.
Day 41 Python EDA	Data loading, Data validation, Cleaning, EDA, Outlier analysis, Trends, Correlation.	Profile patient wait times, correlate age brackets with appointment compliance, and perform hypothesis tests on no-show rates.	**Healthcare EDA Notebook** Documented Jupyter Notebook (`.ipynb`) with statistical tests.
Day 42 Power BI BI	Data model, Relationships, DAX measures, Operations & Financial dashboards.	Build Star Schema model. Create measures: Bed Occupancy %, Cancellation Rate %, Revenue per Bed. Design 3-page cockpit.	**Healthcare BI Dashboard** Complete interactive Power BI Dashboard file (`.pbix`).
Day 43 Presentation	Root-cause analysis, Meaningful insights, Business impact, Strategic recommendations, Executive storytelling.	Present: Problem → Analysis → Findings → Root Cause → Recommendation → Expected Impact.	**Executive Board Presentation** 10-slide deck + Oral Viva defense before evaluation committee.

PHASE 7: PORTFOLIO ENGINEERING & CAREER READINESS (DAY 44) Industry Launchpad

Day 44: Professional Data Analyst Portfolio & GitHub Master Showcase

Core Topics: GitHub basics, Repository creation, Markdown README files, Project documentation, Dashboard screenshots, Storytelling, Resume descriptions, LinkedIn portfolio.
Target Candidate Portfolio: 5 Major Capstone Projects (Excel Dashboard, SQL E-Commerce Engine, A/B Testing Econometrics, Python Customer Behavior, Healthcare Intelligence Ecosystem) + **15+ Verified Mini-Projects**.
Deliverable: Live public GitHub Repository URL + Industry-grade Technical Resume & LinkedIn Profile.

4. Academic Lab Specifications & Ethical Guidelines

Recommended Enterprise IT Lab Infrastructure:

- Hardware: Dual-Core Intel i3/i5 or AMD Ryzen 3/5, 8GB RAM, 100GB Free Disk
- Operating System: Windows 10/11 64-bit or Ubuntu 22.04 LTS
- Software: MS Excel / LibreOffice Calc, MySQL 8.0 / SQLite3
- Python: Python 3.10+ (Jupyter Notebook / Google Colab for low-spec PCs)
- BI: Power BI Desktop (Free Academic Edition) + Git & GitHub Desktop

Data Privacy & Academic Ethics (DPDP Act 2023):

- All project implementations strictly utilize anonymized or enterprise synthetic data.
- Strict zero-tolerance policy against plagiarism in SQL queries and code.
- Ethical analytics mandates that dashboards must avoid demographic biases.
- Career & Higher Ed Pathways: Data Analyst, BI Specialist, B.Tech/MCA Analytics Tracks.